

第7章 测量和数据采集系统

7.1 电量测量（自学）

7.2 非电量电测法和数据采集系统的组成（自学）

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7.5 模拟开关和取样-保持电路（自学）

7.6 数-模（D/A）转换器

7.7 模-数（A/D）转换器

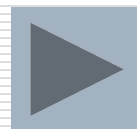
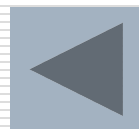
7.8 非电量测量系统举例（自学）



7.4 有源滤波器和测量放大电路

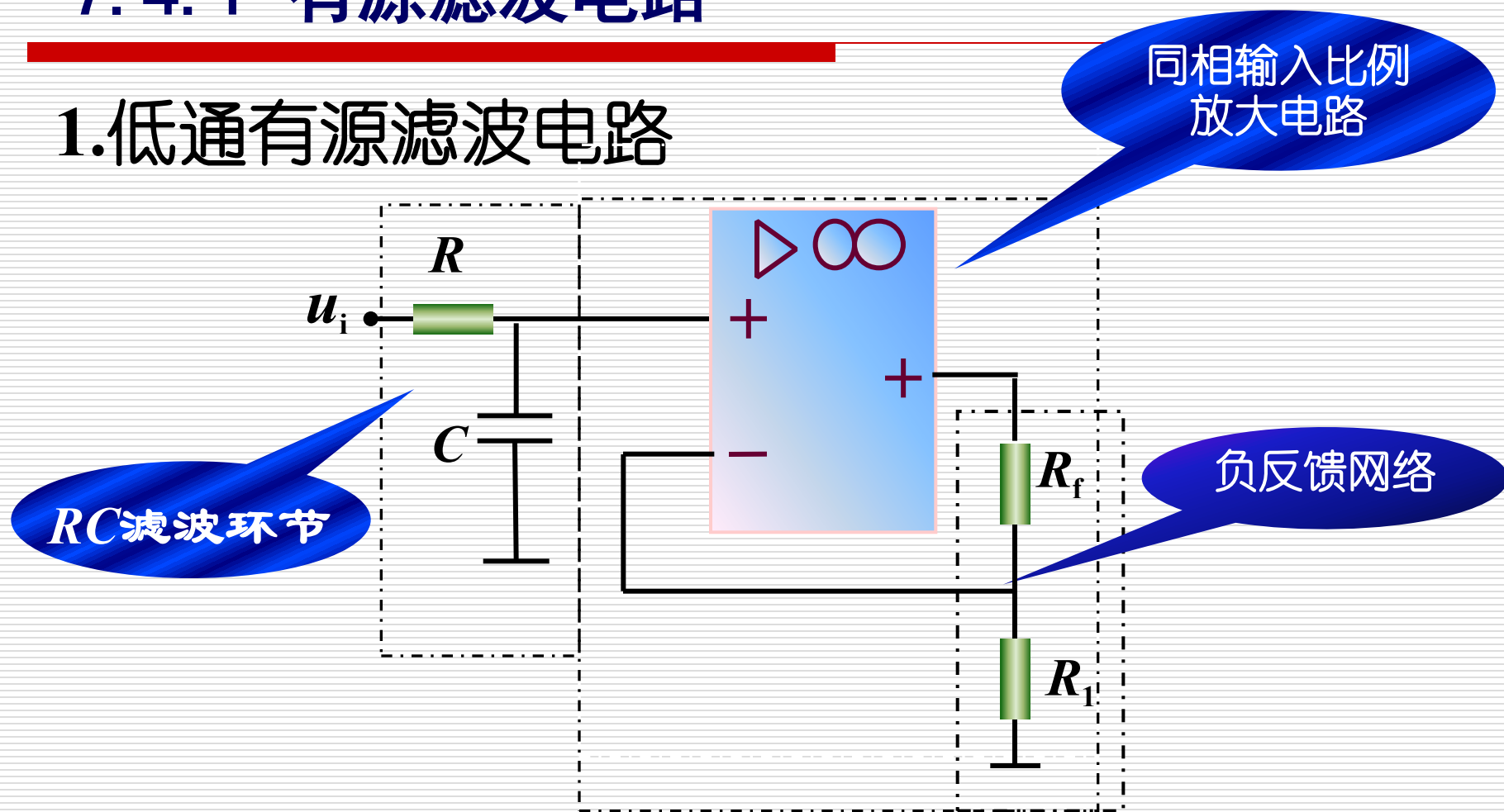
7.4.1 有源滤波电路

7.4.2 测量放大电路

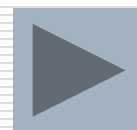
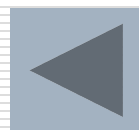


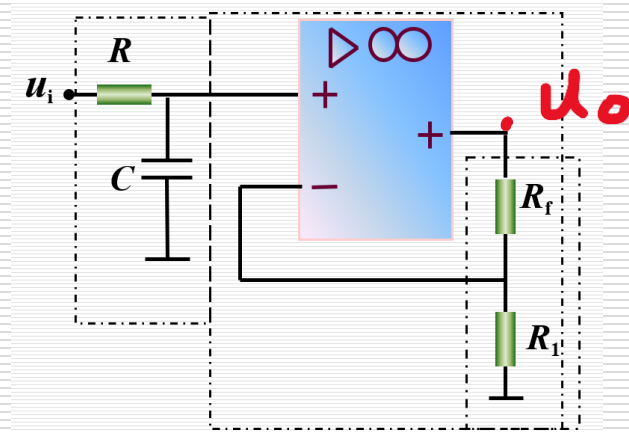
7.4.1 有源滤波电路

1. 低通有源滤波电路



一阶低通有源滤波电路





$$\dot{U}_o = \left(1 + \frac{R_f}{R_1}\right) \frac{-j\frac{1}{\omega C}}{R - j\frac{1}{\omega C}} \dot{U}_i = \left(1 + \frac{R_f}{R_1}\right) \frac{\dot{U}_i}{1 + j\omega RC} = A_f \frac{\dot{U}_i}{1 + j\frac{\omega}{\omega_c}}$$

$$H(j\omega) = \frac{\dot{U}_o}{\dot{U}_i} = A_f \frac{1}{1 + j\frac{\omega}{\omega_c}} = \frac{A_f}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^2}} \angle -\arctan\left(\frac{\omega}{\omega_c}\right) \quad \omega_c = \frac{1}{RC}$$

$$|H(j\omega)| = \frac{A_f}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^2}}$$

—— 幅频特性

$$\varphi(\omega) = -\arctan\left(\frac{\omega}{\omega_c}\right)$$

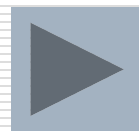
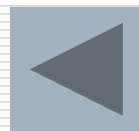
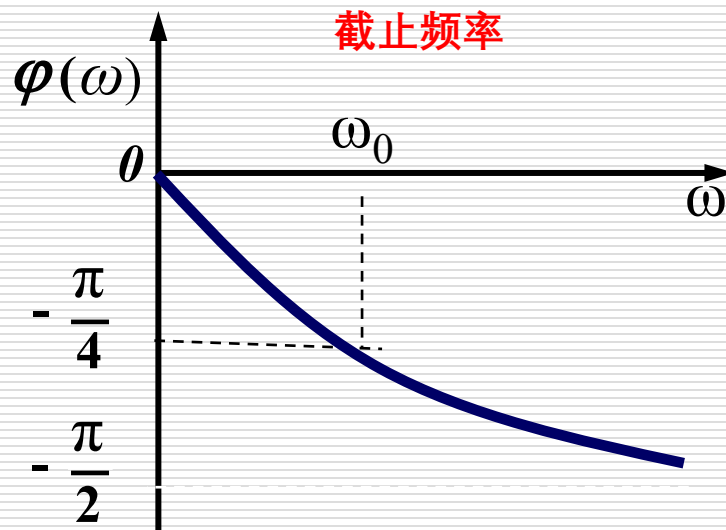
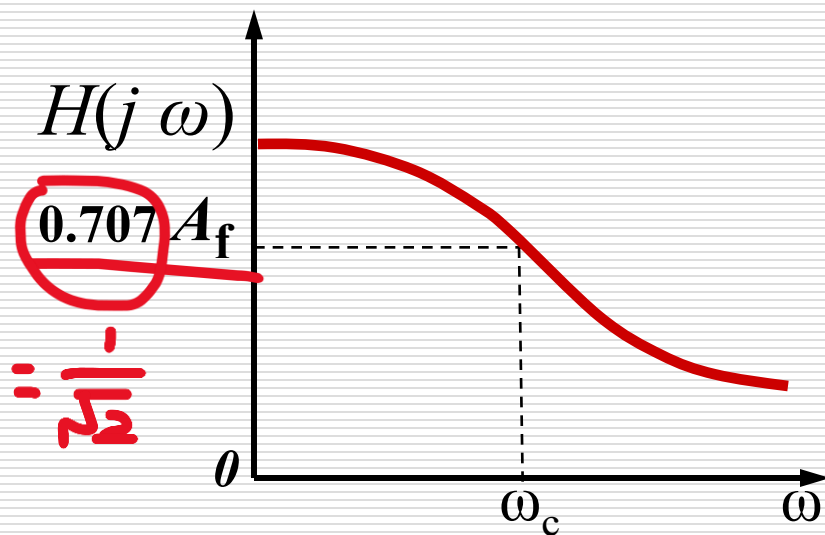
—— 相频特性

$$|H(j\omega)| = \frac{A_f}{\sqrt{1 + \left(\frac{\omega}{\omega_c}\right)^2}}$$

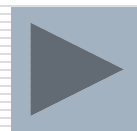
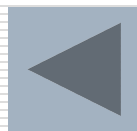
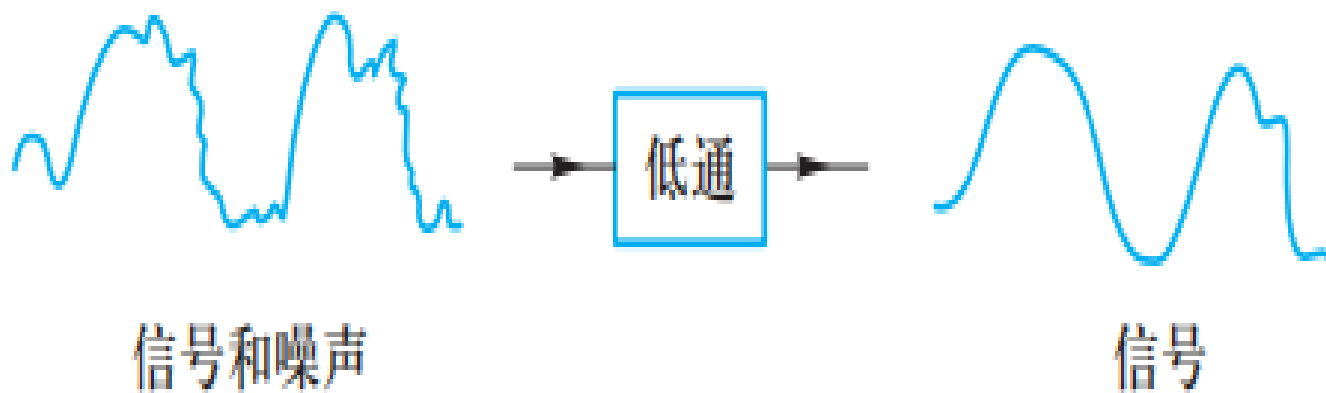
—— 幅频特性

$$\varphi(\omega) = -\arctan\left(\frac{\omega}{\omega_c}\right)$$

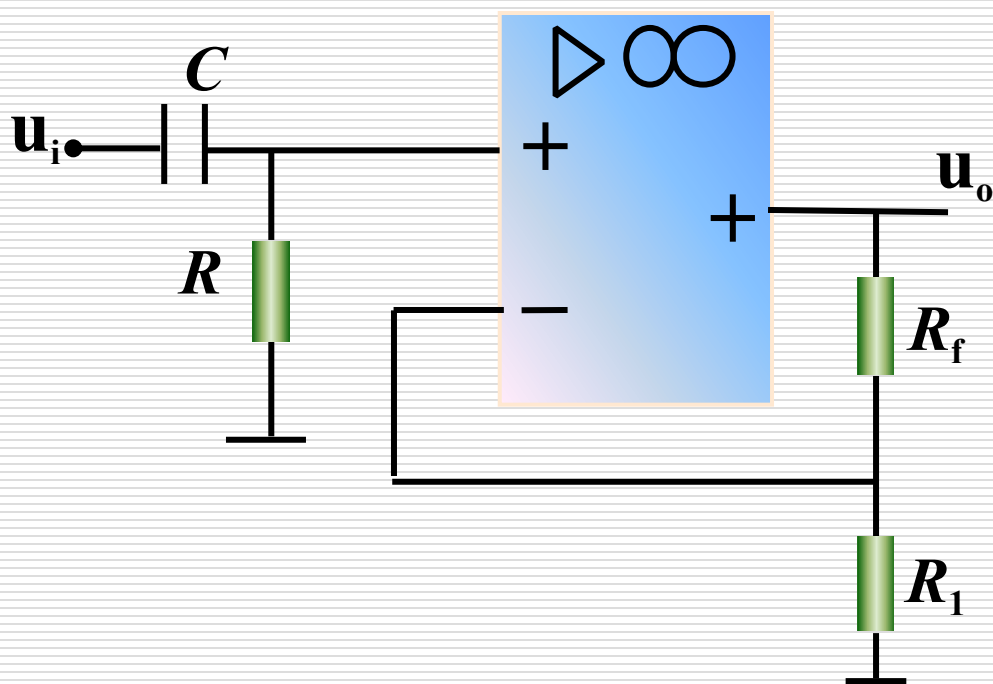
—— 相频特性



用低通滤波器进行信号处理示例



高通滤波电路



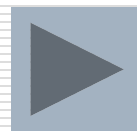
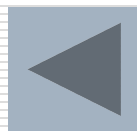
$$\dot{U}_O = \left(1 + \frac{R_f}{R_1} \right) \frac{R}{R + j \frac{1}{\omega C}} \dot{U}_i$$

$$= \frac{A_f R}{R + j \frac{1}{\omega C}} \dot{U}_i$$

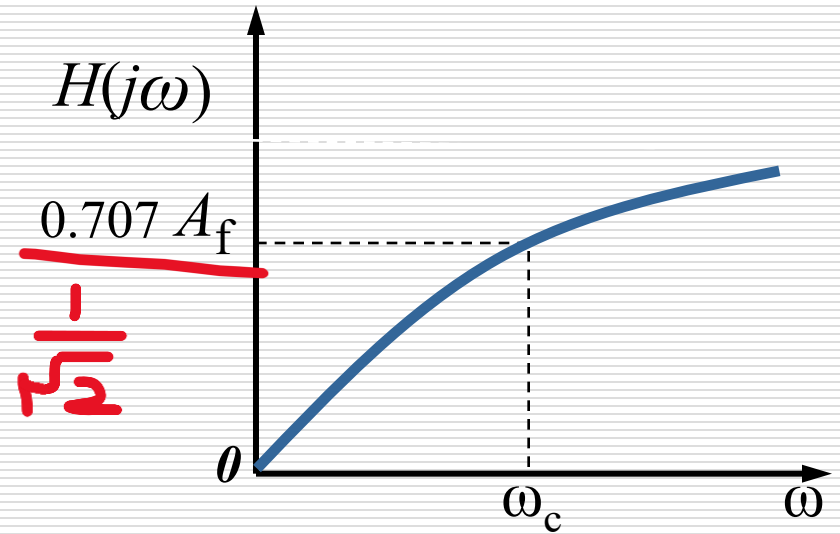
$$= \frac{A_f}{1 - j \frac{\omega C}{\omega}} \dot{U}_i$$

$$\text{其中 } \omega_c = \frac{1}{RC}$$

一阶高通滤波电路

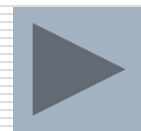
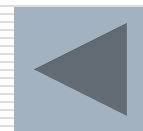
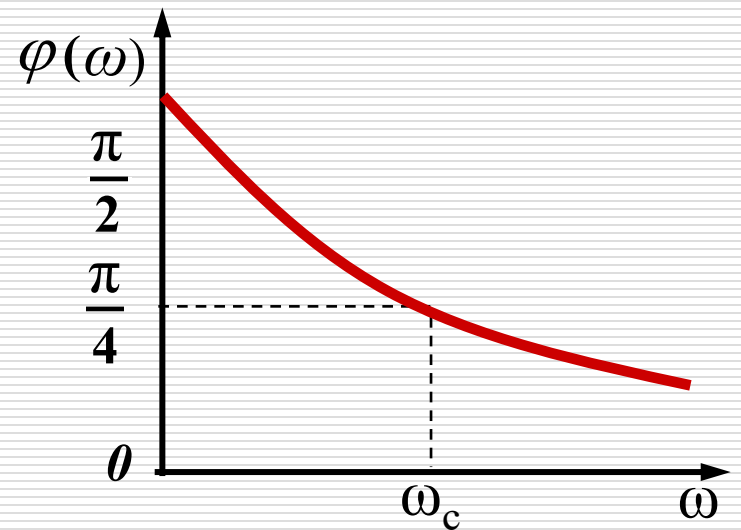


$$H(j\omega) = \frac{\dot{U}_o}{\dot{U}_i} = A_f \frac{1}{1 - j \frac{\omega_c}{\omega}}$$

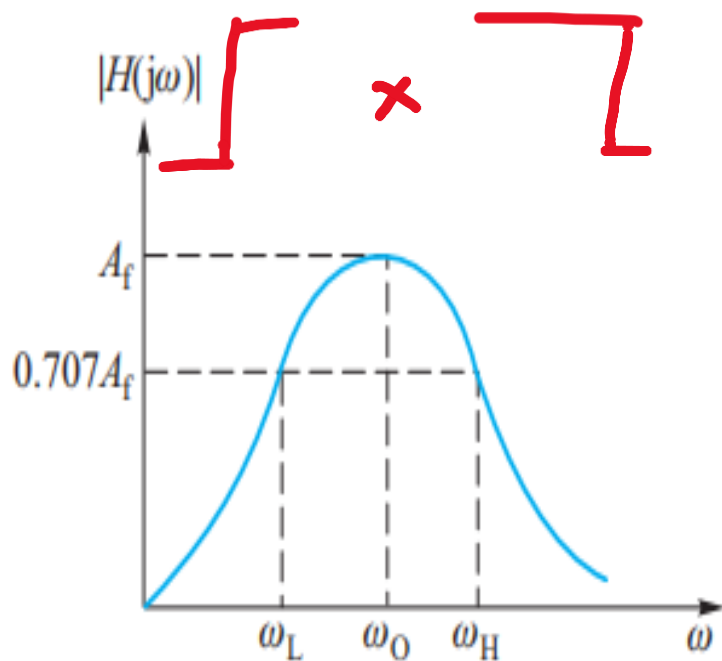


$$|H(j\omega)| = \frac{A_f}{\sqrt{1 + \left(\frac{\omega_c}{\omega}\right)^2}}$$

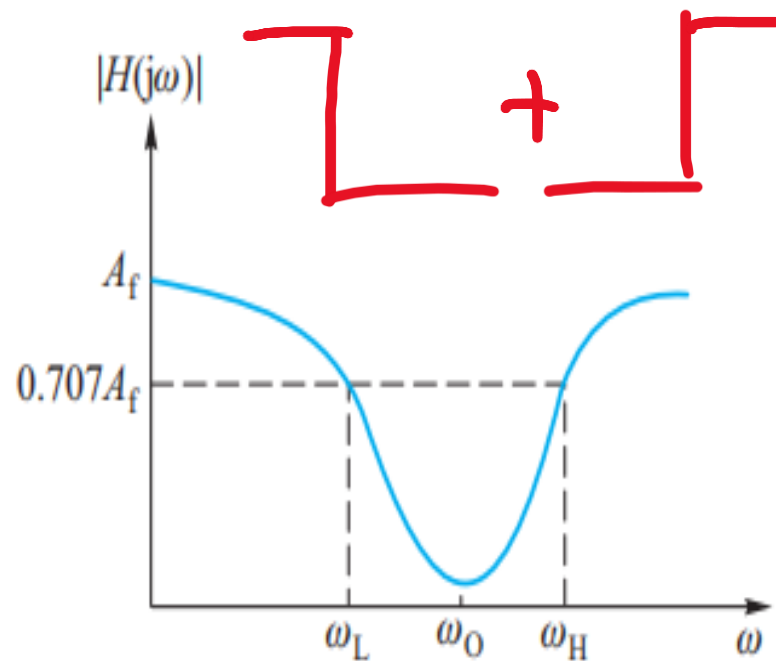
$$\varphi(\omega) = -\arctan\left(\frac{\omega_c}{\omega}\right)$$



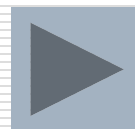
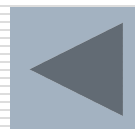
带通和带阻滤波电路的幅频特性



(a)



(b)



7.4.2 测量放大电路



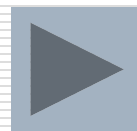
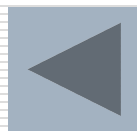
测量放大电路的作用

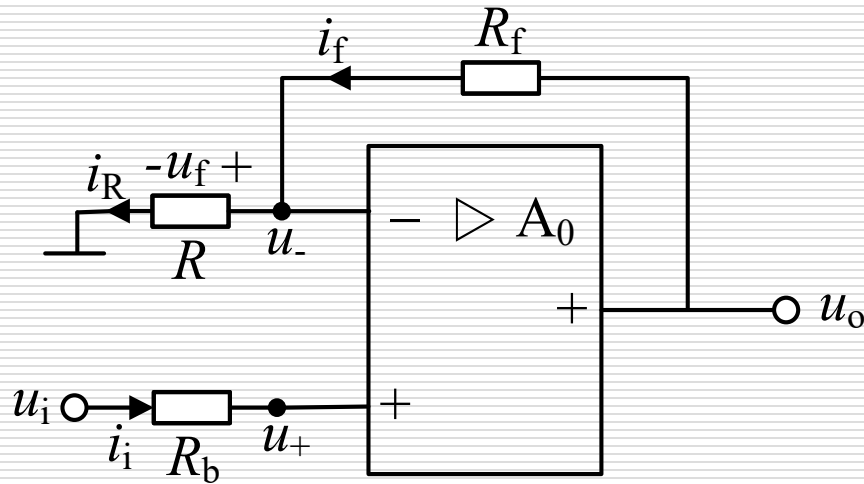
将传感器送来的微弱信号进行放大，再送到后面电路去处理。



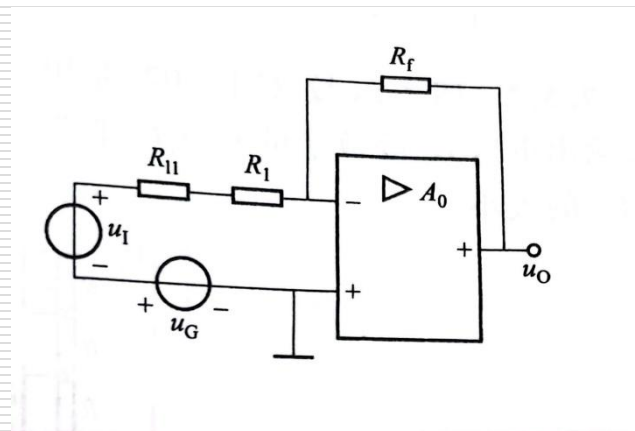
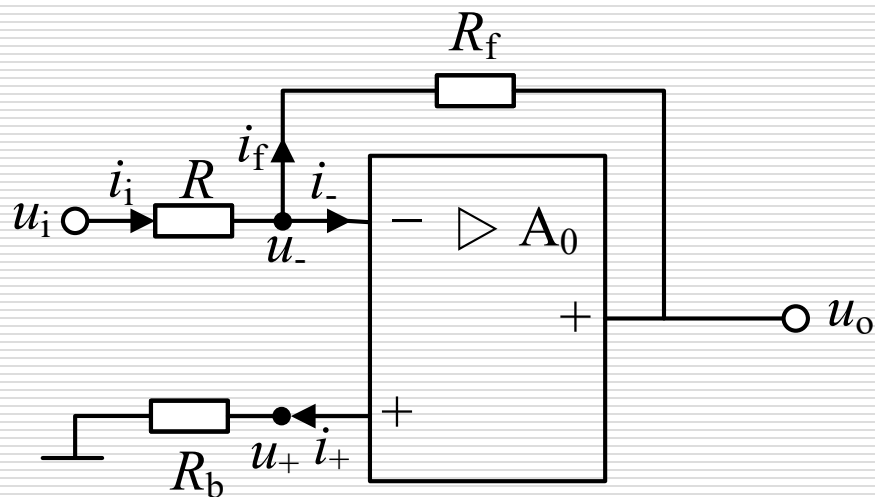
对测量放大电路的要求

输入电阻高、噪声低、稳定性好、精度及可靠性高、共模抑制比大、线性度好、失调小、并有一定的抗干扰能力。





同相输入放大电路**输入电阻高**，但在集成运放的两个输入端有**共模**信号加入，对环境的共模信号干扰比较敏感。采用这种方式的电路，要选用共模输入电压范围大，共模抑制比高的集成运放，并在电路上采取必要措施滤除外部的共模信号。



$$u_o = -\frac{R_f}{R_1 + R_{L1}}(u_I + u_G)$$

- (1) 反相输入放大电路不必考虑运放本身的共模抑制比问题。
- (2) 远距离测量时，由于地电阻会引入干扰，或由于传感器的工作环境恶劣，造成在传感器输出端产生干扰，这些干扰信号被放大后输出，严重影响电路的性能。
- (3) 输入电阻较低，不宜与传感器直接连接。

例:7.4.5 差分输入运算放大器广泛应用于测量仪表放大器的输入级,如图 7.05 所示。图中导线电阻 $R_{L1}=R_{L2}=1\ \Omega$,信号地(c端)和仪表地(d端)之间的地电阻 R_G 上存在干扰电压 $u_G=0.1\ \text{V}$ 。试求:(1) 根据图示数据,输出电压 u_O 为多大?(提示:此时 a 端和 d 端(仪表地)之间的电压 $u_{ad}=u_1+u_G$,b 和 d 之间的电压 $u_{bd}=u_G$ 。)(2) 如果改用反相输入运算放大器(也就是在图 7.05 中不接 R_{L2} 和 R_2 且 $R_3=0$),输出电压 u_O 将是多大?

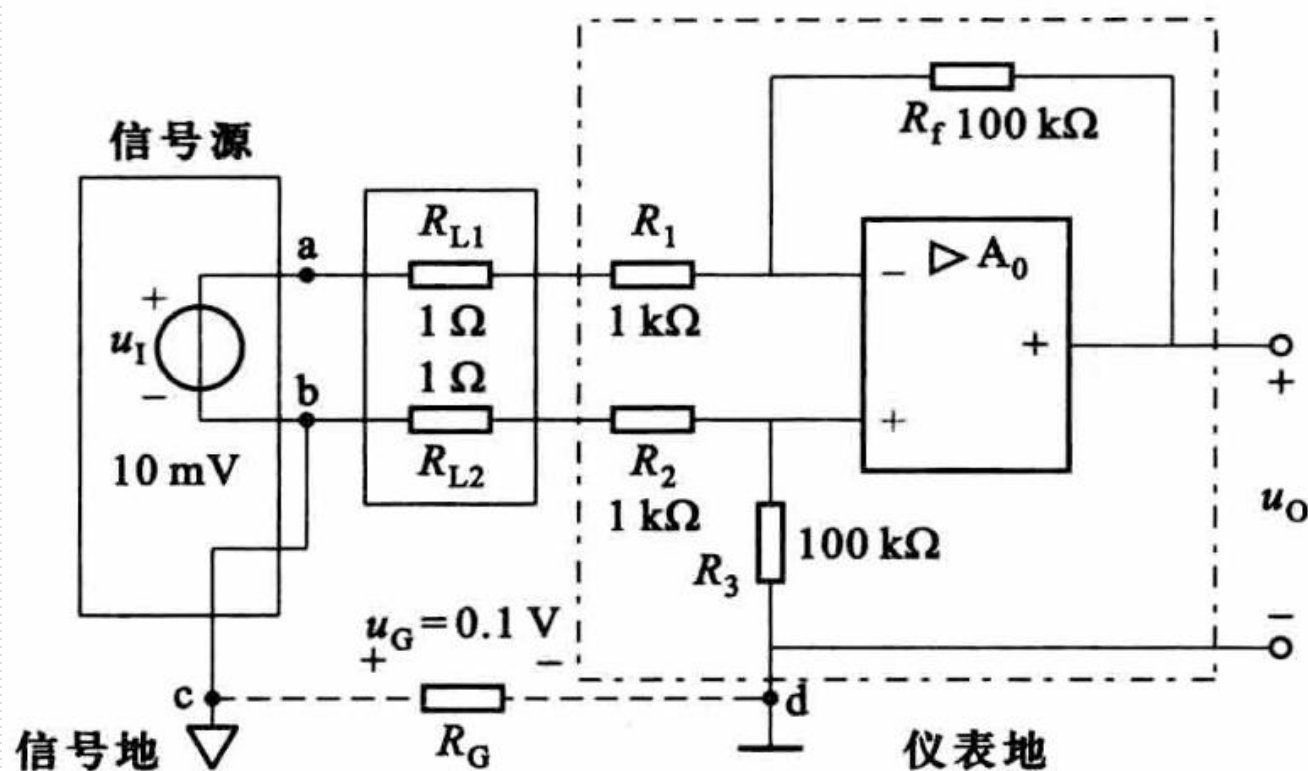
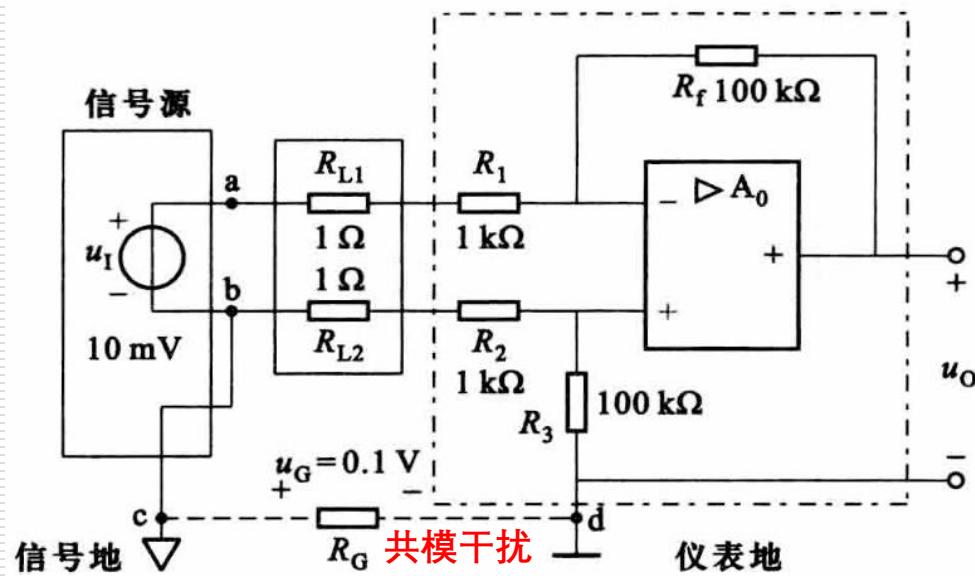
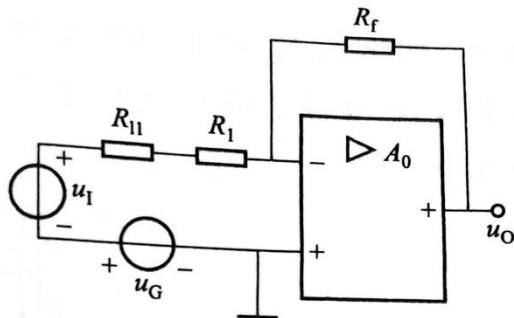


图 7.05 习题 7.4.5 的电路

例:

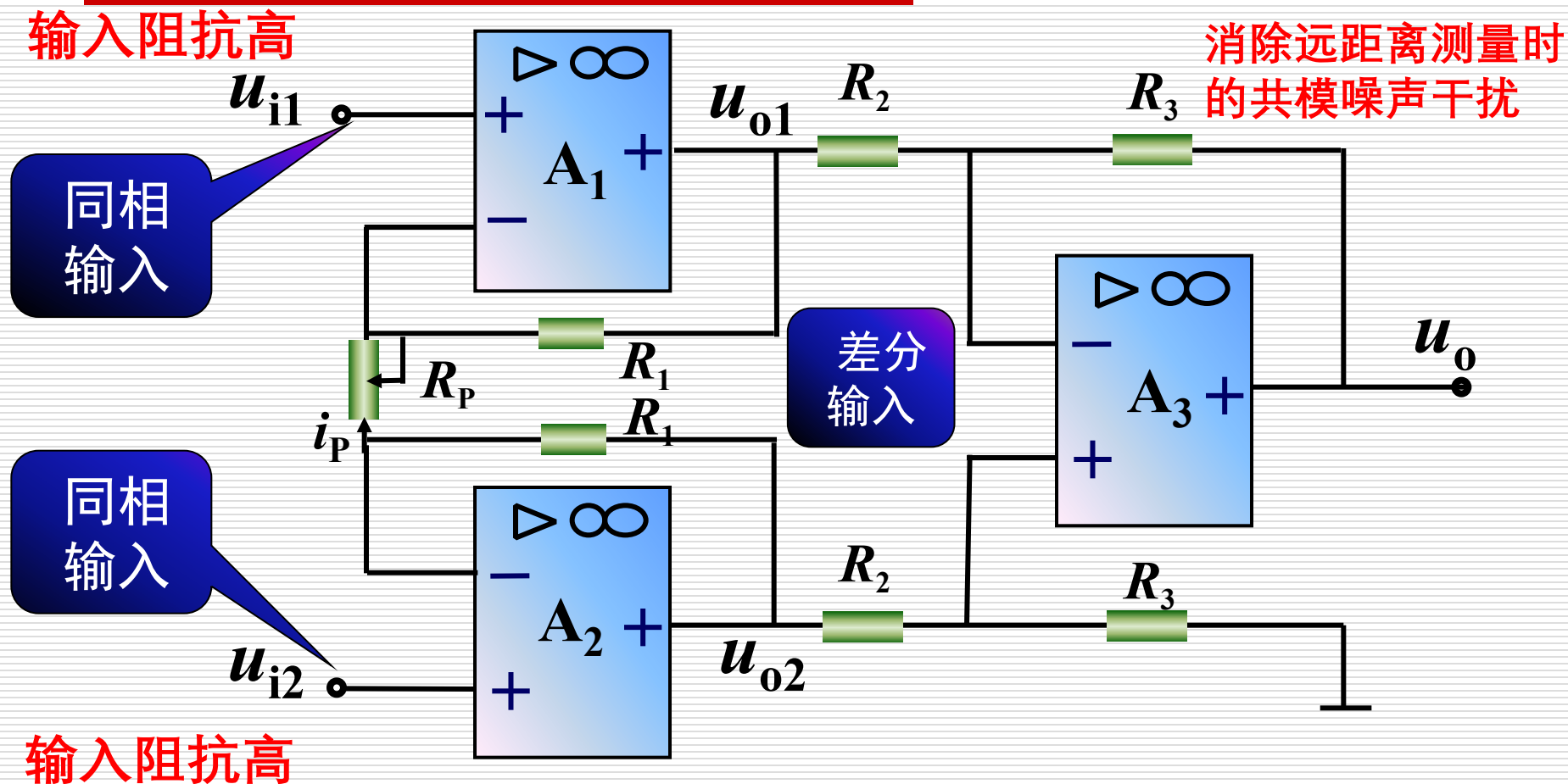


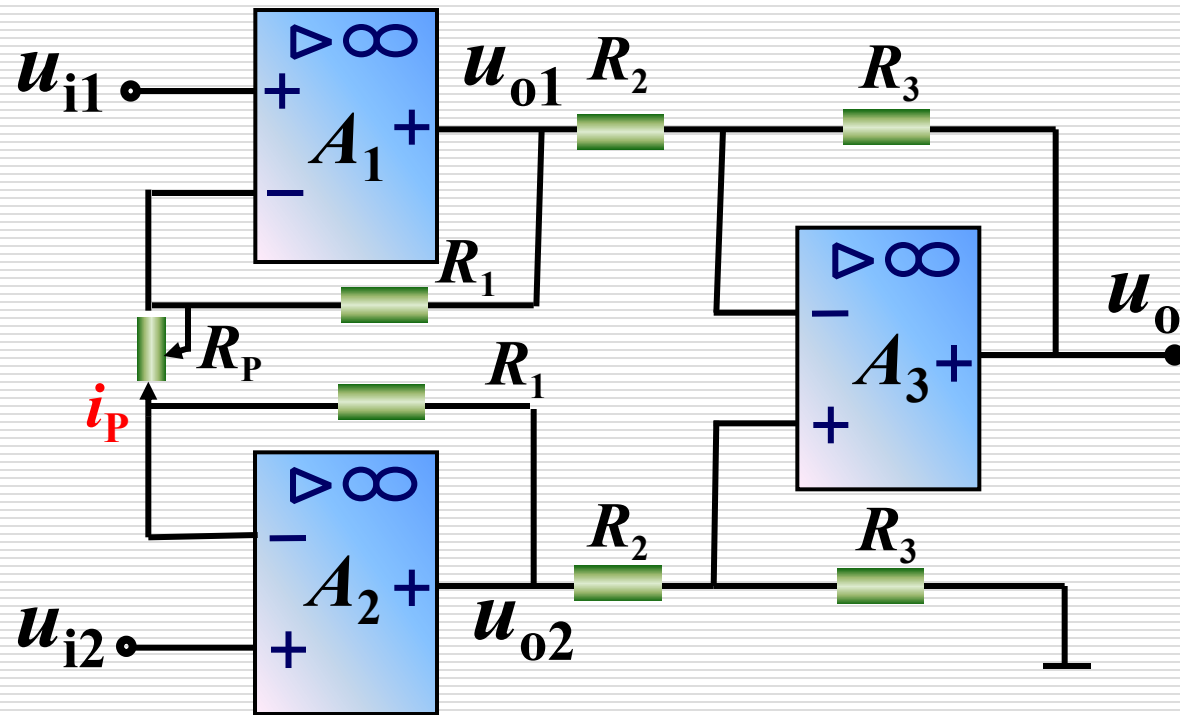
$$u_o = \frac{R_f}{R_1 + R_{L1}} (u_{bd} - u_{ad}) = -\frac{R_f}{R_1 + R_{L1}} u_I$$



$$u_o = -\frac{R_f}{R_1 + R_{L1}} (u_I + u_G)$$

1. 用集成运放组成的测量放大电路





$$i_P = \frac{u_{i2} - u_{i1}}{R_P}$$

$$A_d = \frac{u_o}{u_{i2} - u_{i1}}$$

$$= \frac{R_3}{R_2} \left(1 + \frac{2R_1}{R_P} \right)$$

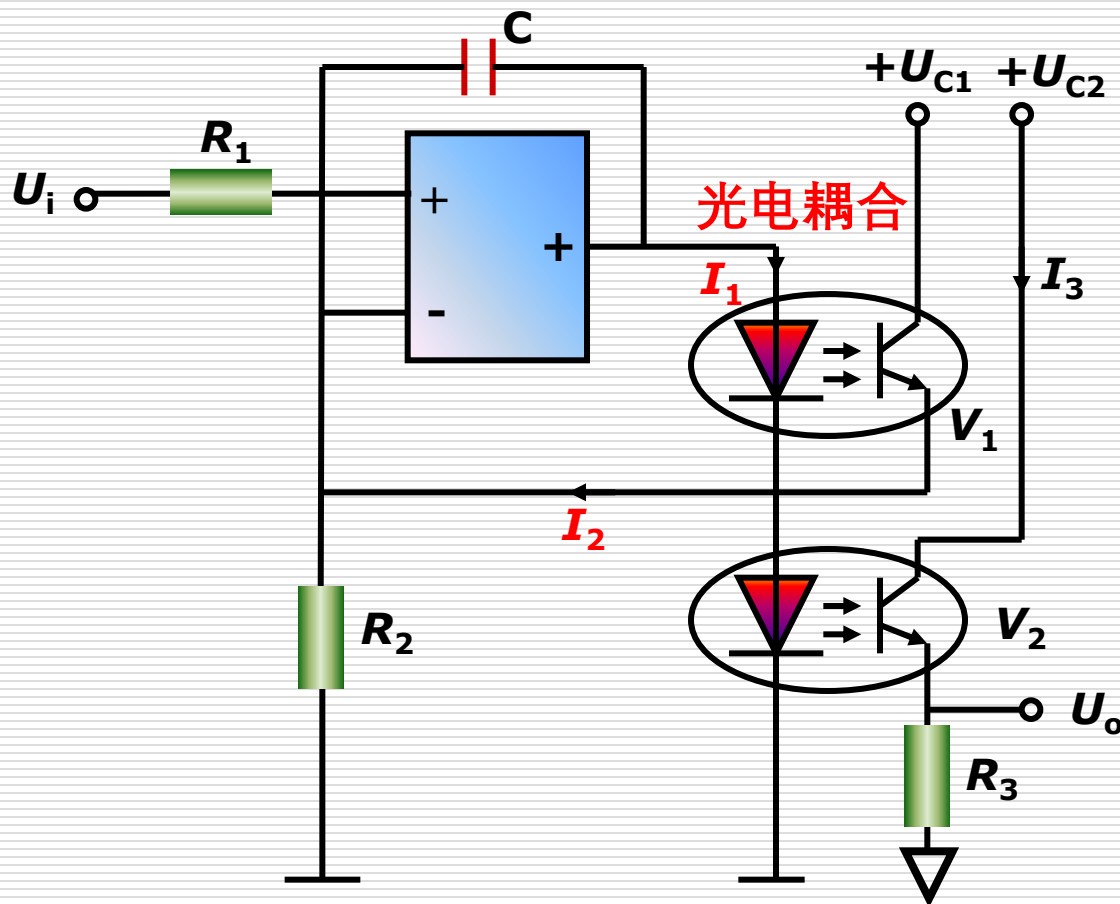
$$u_{o2} - u_{o1} = (R_1 + R_P + R_1) i_P = \left(1 + \frac{2R_1}{R_P} \right) (u_{i2} - u_{i1})$$

$$u_o = \frac{R_3}{R_2} (u_{o2} - u_{o1}) = \frac{R_3}{R_2} \left(1 + \frac{2R_1}{R_P} \right) (u_{i2} - u_{i1})$$



3 隔离放大器

消除自激振荡



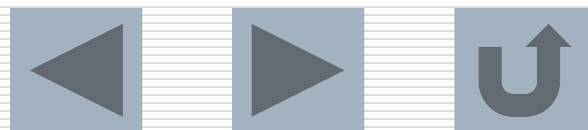
$$I_2 = g_1(I_1)$$

$$I_3 = g_2(I_1)$$

$$\begin{aligned} A &= \frac{U_o}{U_i} = \frac{R_3 I_3}{R_2 I_2} \\ &= \frac{R_3}{R_2} \cdot \frac{g_2(I_1)}{g_1(I_1)} \\ &= \frac{R_3}{R_2} \end{aligned}$$

若特性相同

不共地



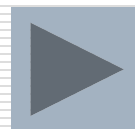
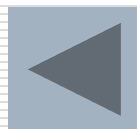
7.5 模拟开关和取样-保持电路

7.5.1 模拟开关

7.5.2 取样-保持 (S/H) 电路

为了使得取样保持电路输出的信号能不失真地复现为原输入信号，必须满足

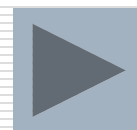
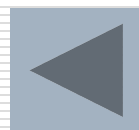
$$f_s \geq f_{\text{imax}}$$



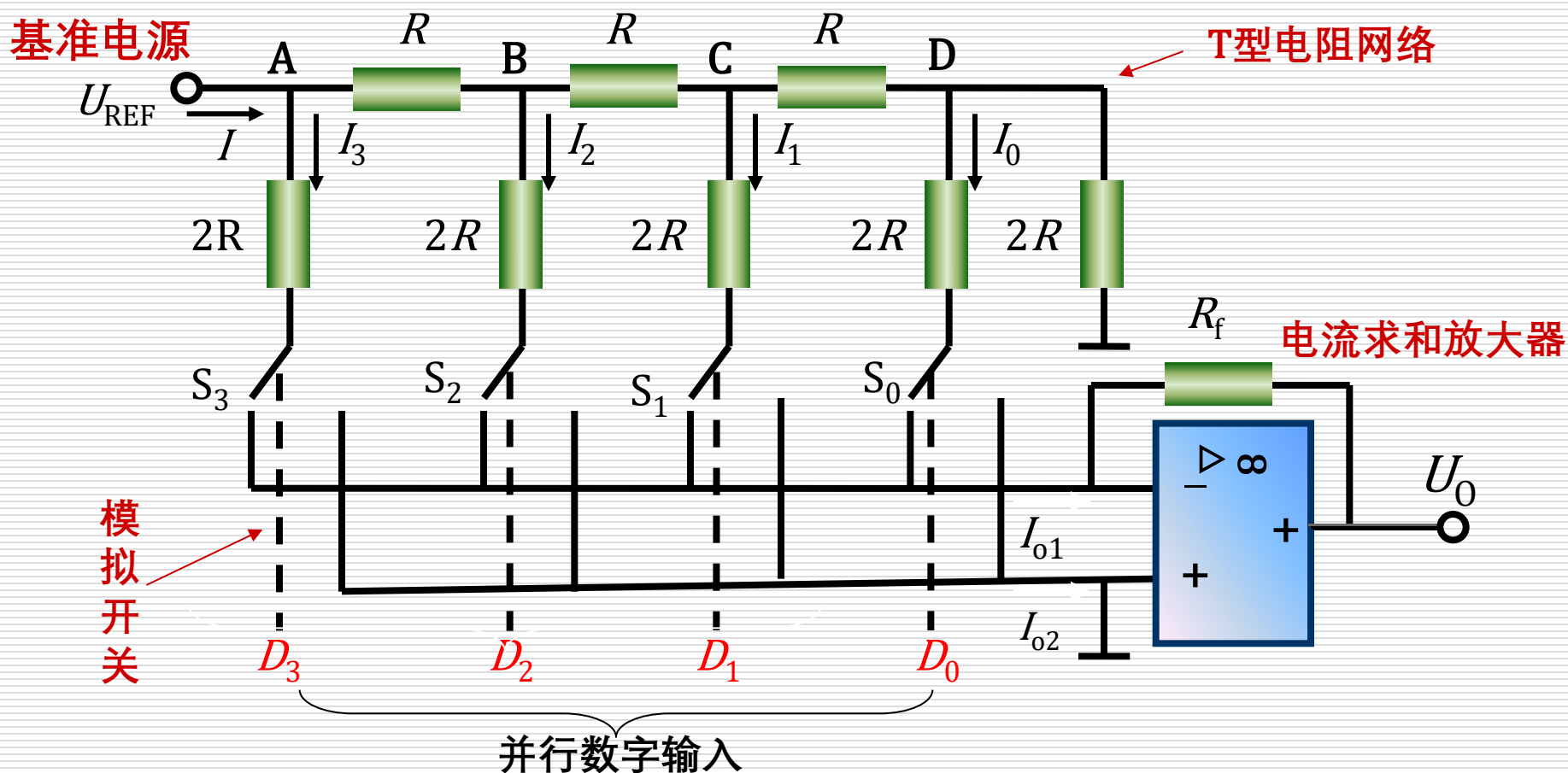
7.6 数/模(D/A)转换器

7.6.1 T型电阻网络D/A转换器

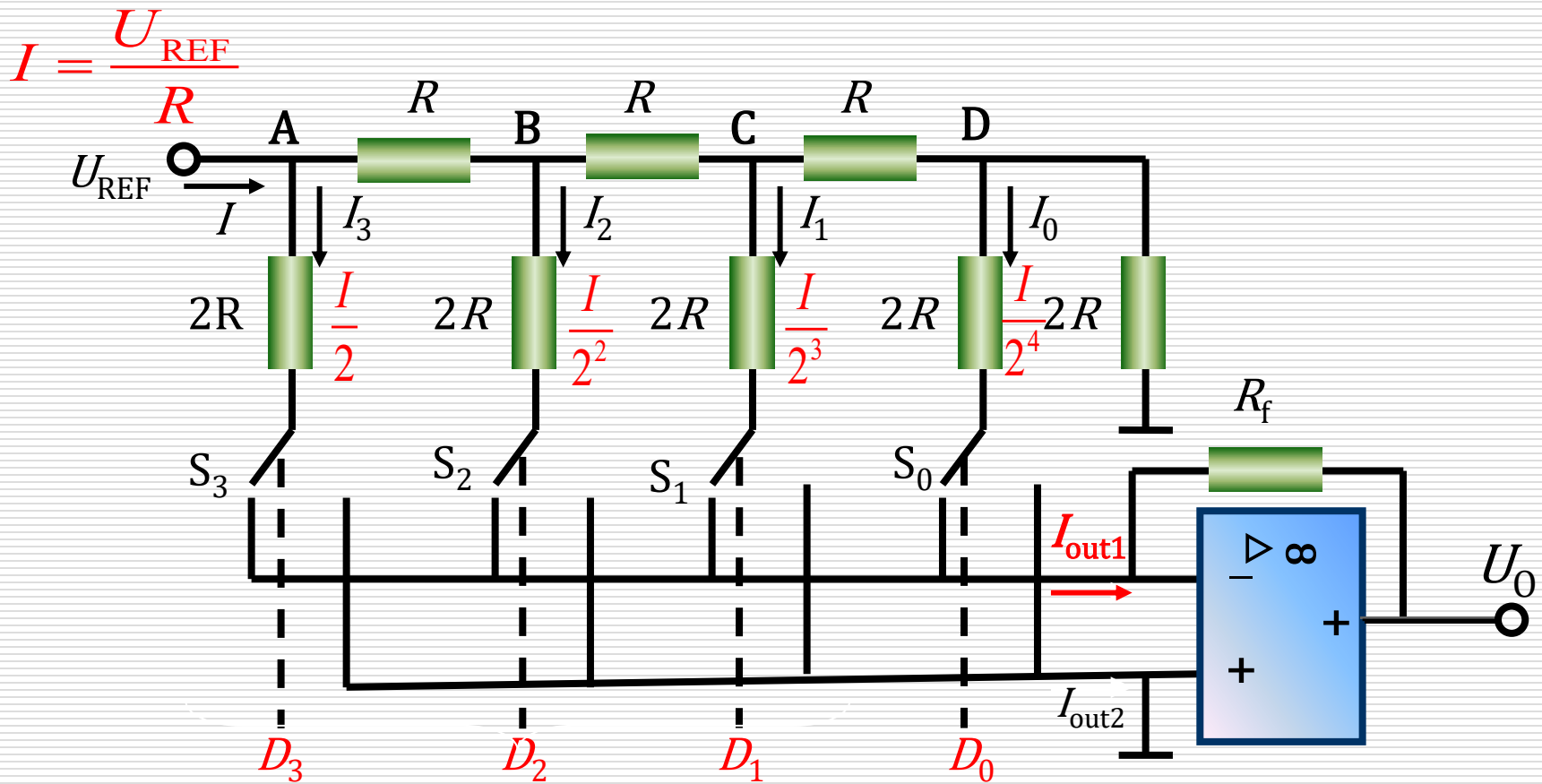
7.6.2 集成D/A转换器举例



7.6.1 T型电阻网络D/A转换器



$D=0$, S 接地（打到右侧）； $D=1$, S 接“虚地”（打到左侧）。无论打到哪边，都是“地”。



$$I_{\text{out1}} = \frac{I}{2} D_3 + \frac{I}{2^2} D_2 + \frac{I}{2^3} D_1 + \frac{I}{2^4} D_0 = \frac{I}{2^4} (D_3 2^3 + D_2 2^2 + D_1 2^1 + D_0 2^0)$$

$$\begin{aligned} U_o &= -R_f I_{\text{out1}} = -\frac{R_f I}{2^4} (D_3 2^3 + D_2 2^2 + D_1 2^1 + D_0 2^0) \\ &= -\frac{U_{\text{REF}} R_f}{2^4 R} (D_3 2^3 + D_2 2^2 + D_1 2^1 + D_0 2^0) \end{aligned}$$

$$\begin{aligned}
 U_o &= -R_f I_{\text{out1}} = -\frac{R_f I}{2^4} (D_3 2^3 + D_2 2^2 + D_1 2^1 + D_0 2^0) \\
 &= -\frac{U_{\text{REF}} R_f}{2^4 R} (\textcolor{red}{D}_3 2^3 + \textcolor{red}{D}_2 2^2 + \textcolor{red}{D}_1 2^1 + \textcolor{red}{D}_0 2^0)
 \end{aligned}$$

对 n 位D/A有

$$U_o = -\frac{U_{\text{REF}} R_f}{2^n R} (\textcolor{red}{D}_{n-1} 2^{n-1} + \textcolor{red}{D}_{n-2} 2^{n-2} + \dots + \textcolor{red}{D}_0 2^0)$$

$$U_{\text{omax}} = -\frac{U_{\text{REF}} R_f}{2^n R} (2^{n-1} + 2^{n-2} + \dots + 2^0) = -\frac{U_{\text{REF}} R_f}{2^n R} \left(\frac{2^0 - 2^n}{1 - 2} \right) = -\frac{U_{\text{REF}} R_f}{2^n R} (2^n - 1) = -\frac{U_{\text{REF}} R_f}{R} \left(\frac{\textcolor{red}{2}^n - \textcolor{red}{1}}{\textcolor{red}{2}^n} \right)$$

$$U_{\text{omin}} = -\frac{U_{\text{REF}} R_f}{2^n R} (0 + 0 + \dots + 2^0) = -\frac{U_{\text{REF}} R_f}{2^n R} = -\frac{U_{\text{REF}} R_f}{R} \frac{\textcolor{red}{1}}{\textcolor{red}{2}^n}$$

$$\text{定义分辨率} = \frac{U_{\text{omin}}}{U_{\text{omax}}} = \frac{1}{2^n - 1}$$

D/A的主要技术参数

※ 分辨率:

是指最小输出电压（对应的输入二进制数最低位为1）与最大输出电压(对应的输入二进制数的所有位全为1)之比。

$$\text{分辨率} = 1 / (2^n - 1)$$

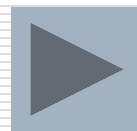
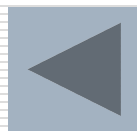
例如十位数模转换器的分辨率为: $\frac{1}{2^{10}-1} \approx 0.001$

※ 转换精度:

表示实际输出的电压值与理想的输出电压值之间的差别。

※ 转换速度:

从数码输入到模拟电压稳定输出间的时间称为转换速度。



$$U_{\text{omax}} = -\frac{U_{\text{REF}} R_f}{R} \left(\frac{2^n - 1}{2^n} \right) \quad U_{\text{omin}} = -\frac{U_{\text{REF}} R_f}{R} \frac{1}{2^n} \quad \frac{U_{\text{omin}}}{U_{\text{omax}}} = \frac{1}{2^n - 1}$$

例：若8位D/A转换器最大输出电压是9.945V，则当输入为10100101时，输出电压是多少？

$$\frac{U_{\text{omin}}}{U_{\text{omax}}} = \frac{1}{2^n - 1} \Rightarrow U_{\text{omin}} = \frac{U_{\text{omax}}}{2^n - 1}$$

10100101对应的十进制数为 $2^7 + 2^5 + 2^2 + 2^0 = 128 + 32 + 4 + 1 = 165$

故输出电压为：

$$165 \cdot U_{\text{omin}} = 165 \cdot \frac{U_{\text{omax}}}{2^n - 1} = 165 \cdot \frac{9.945}{255} = 6.435\text{V}$$

$$U_{\text{omax}} = -\frac{U_{\text{REF}} R_f}{R} \left(\frac{2^n - 1}{2^n} \right)$$



$$U_{\text{omax}} = U'_{\text{REF}} \left(\frac{2^n - 1}{2^n} \right)$$

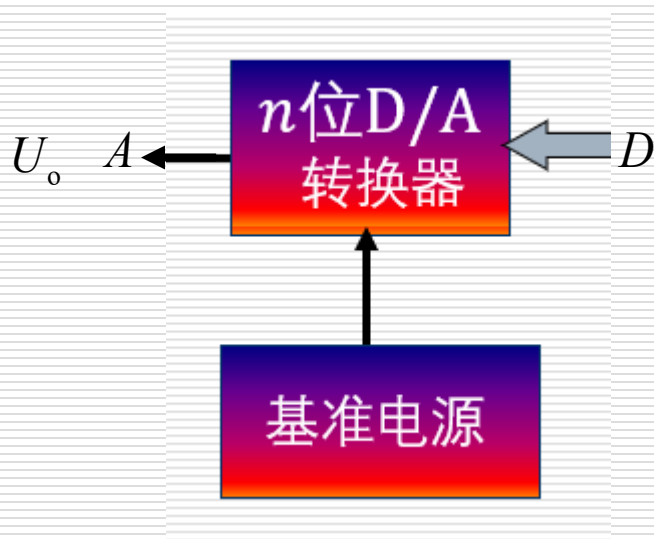
$$U_{\text{omin}} = -\frac{U_{\text{REF}} R_f}{R} \frac{1}{2^n}$$



$$U_{\text{omin}} = U'_{\text{REF}} \frac{1}{2^n}$$

$$\frac{U_{\text{omin}}}{U_{\text{omax}}} = \frac{1}{2^n - 1}$$

若 $n=3$, 有 $U_{\text{omax}} = \frac{7}{8} U'_{\text{REF}}$ $U_{\text{omin}} = \frac{1}{8} U'_{\text{REF}}$



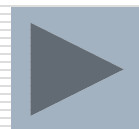
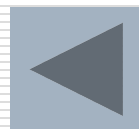
数字量输入			模拟量输出
Q_2	Q_1	Q_0	U_s
1	1	1	$\frac{7}{8} U_{\text{REF}}$
1	1	0	$\frac{6}{8} U_{\text{REF}}$
1	0	1	$\frac{5}{8} U_{\text{REF}}$
1	0	0	$\frac{4}{8} U_{\text{REF}}$
0	1	1	$\frac{3}{8} U_{\text{REF}}$
0	1	0	$\frac{2}{8} U_{\text{REF}}$
0	0	1	$\frac{1}{8} U_{\text{REF}}$
0	0	0	0

7.7 模/数(A/D)转换器

概述

7.7.1 逐次逼近型A/D转换器

*7.7.2 双积分型A/D转换器

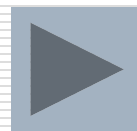
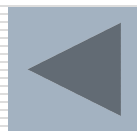


概述

A/D 转换器的作用是将输入的模拟电压数字化。

直接转换器： 逐次逼近型、并联比较型等

间接转换器： 单积分型、双积分型等

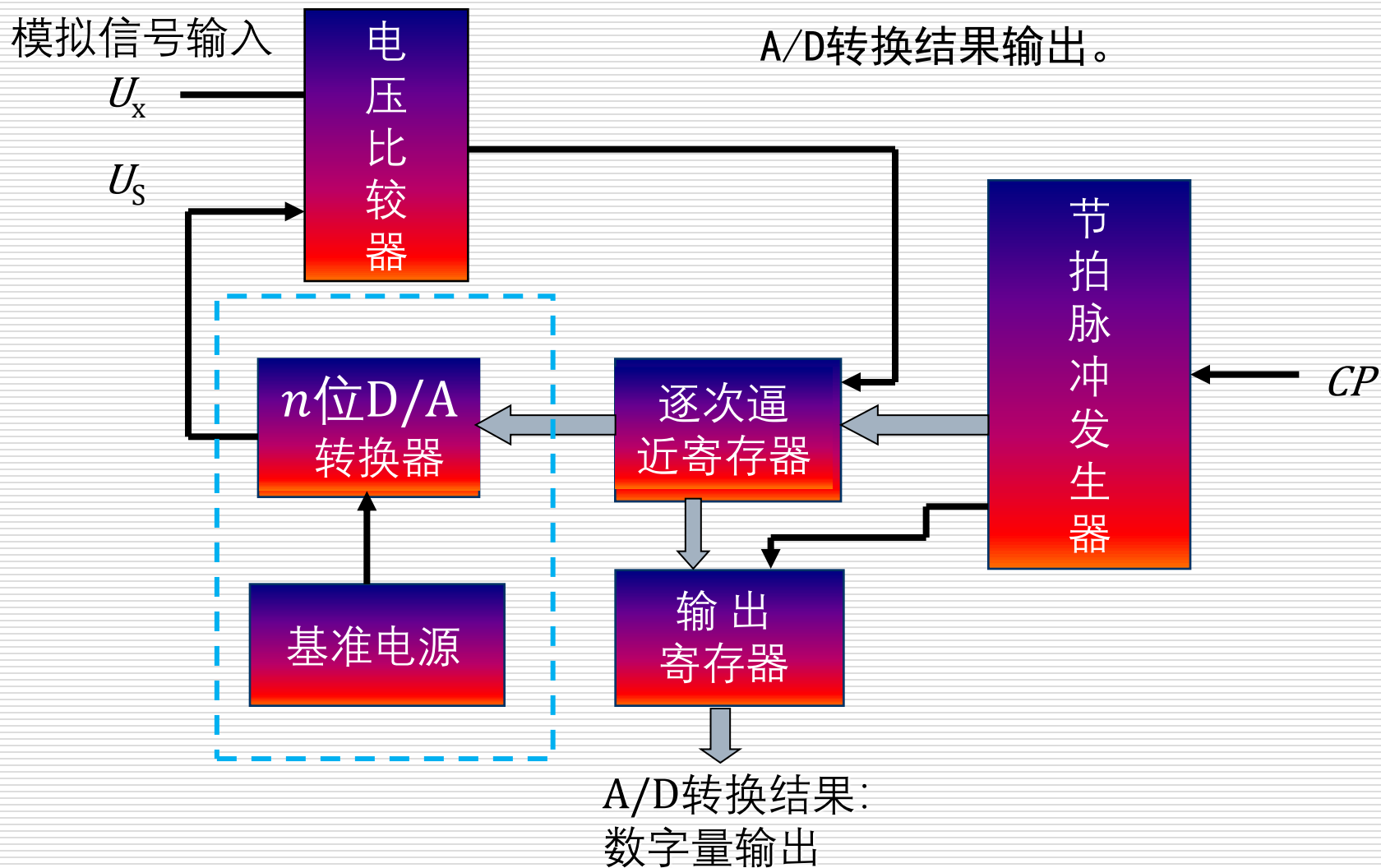


7.7.1 逐次逼近型 A/D转换器

- (1) 逐次逼近寄存器最高位置1，经D/A转换后转换为模拟量 U_s 。
- (2) 与输入信号 U_x 比较决定是否保留。
- (3) 逐次逼近寄存器次高位置1...

...

A/D转换结果输出。

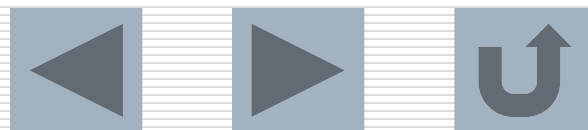
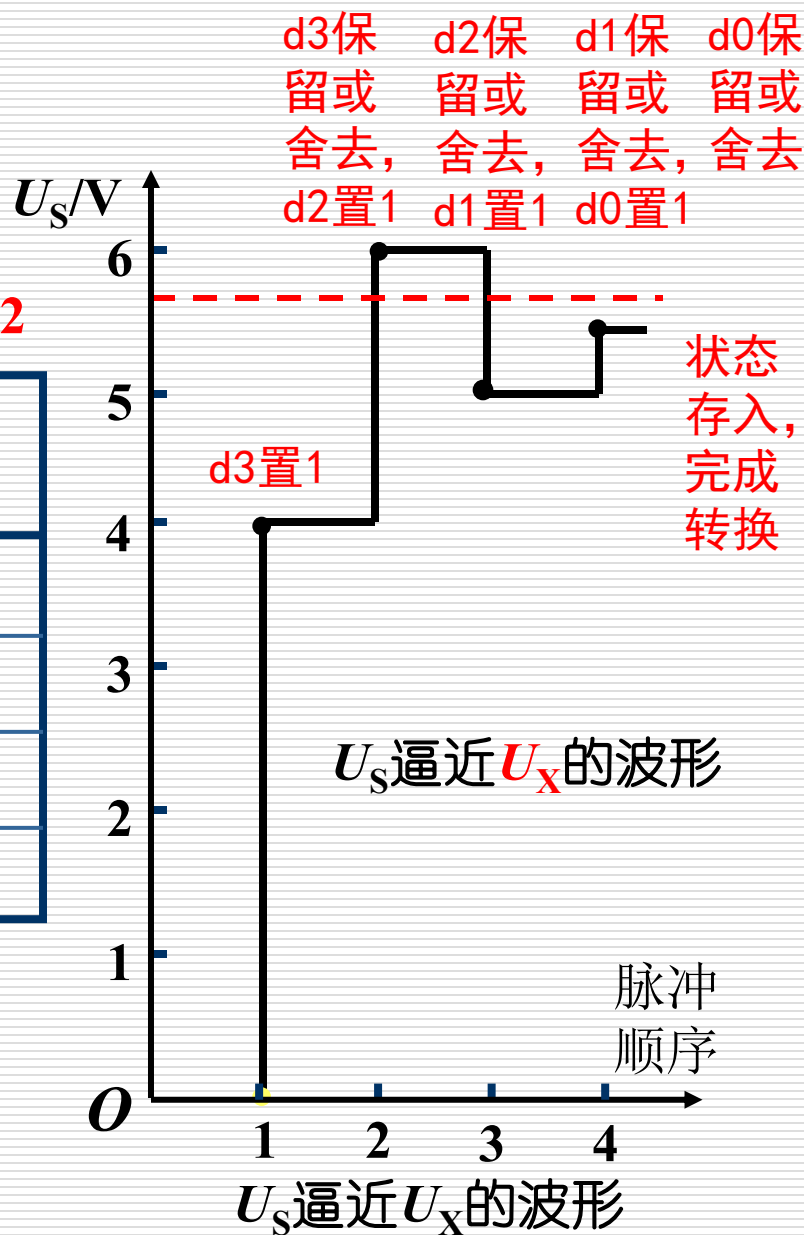


四位逐次逼近型ADC的 转换过程

模拟信号 $U_x = 5.52$

顺序	d_3	d_2	d_1	d_0	U_s/V	比较	该位1的去留
1	1	0	0	0	$4 \frac{8}{16} U'_{REF}$	$U_s < U_x$	留
2	1	1	0	0	$6 \frac{12}{16} U'_{REF}$	$U_s > U_x$	去
3	1	0	1	0	$5 \frac{10}{16} U'_{REF}$	$U_s < U_x$	留
4	1	0	1	1	$5.5 \frac{11}{16} U'_{REF}$	$U_s \approx U_x$	留

对于 n 位 A/D 转换器，转换一次需要 $n+2$ 个时钟脉冲周期。



逐次逼近型A/D转换器精度高，速度快，转换时间固定，容易和微机接口，是目前种类最多，数量最大，应用最广泛的一种A/D转换器件。

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第8章 功率电子电路